ON LOCALIZATION PHENOMENA IN PLASTICITY MODELS

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Abstract.

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1. Introduction

2. The tensile test at small deformations

In this section we develop a one dimensional model for an elastoplastic bar at small deformations. We prove the existence of solutions and we study the boundary value problem that describe tensile test. We show that the model is not able to predict the onset of necking.

2.1. **Multidimensional kinematics.** Let $\Omega = (0,L)\times\Sigma$ with $L>0$ and $\Sigma\subset\mathbb{R}^{d-1}$ be the reference configuration of the cylindrical bar. We denote the generic point in Ω as (x,y) with $x\in(0,L)$ and $y\in\Sigma$. We assume a macroscopic displacement of the form

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} u_1(x) \\ u_2(x)y \end{pmatrix}$$

where u_1 and u_2 are functions from $(0,L)$ to $\mathbb{R}$. This assumption means that the sections of the cylindrical bar remain flat and can translate and expand uniformly. The corresponding displacement gradient is

$$H = \begin{pmatrix} u_1' & 0 \\ u_2' y & u_2 I \end{pmatrix}.$$

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The plastic distortion is assumed to be of the form

$$H_p = \begin{pmatrix} p & 0 \\ 0 & -\frac{p}{d-1} I \end{pmatrix}$$

where $p\colon (0,L) \to \mathbb{R}$. This choice implies plastic incompressibility. The elastic distortion is then given by

$$H_e = H - H_p = \begin{pmatrix} u_1' - p & 0 \\ u_2' y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix}.$$

Then the elastic strain is

$$E_e = \begin{pmatrix} u_1' - p & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & \left(u_2 + \frac{p}{d-1}\right) I \end{pmatrix} = \begin{pmatrix} e_1 & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & e_2 I \end{pmatrix}$$

where we have introduced $e_1 = u_1' - p$ and $e_2 = u_2 + (d-1)^{-1} p$. Assuming an isotropic elastic response the elastic free energy density is given by

$$\psi_e = \frac{1}{2}\kappa (\operatorname{tr} E_e)^2 + \mu |\operatorname{dev} E_e|^2$$

where $\kappa \geq 0$ and $\mu \geq 0$ are the bulk and shear modulus respectively. We notice that

$$\operatorname{tr} E_e = e_1 + (d-1) e_2 \quad \text{and} \quad \operatorname{dev} E_e = \begin{pmatrix} \frac{d-1}{d}(e_1 - e_2) & \frac{u_2'}{2} y^T \\ \frac{u_2'}{2} y & -\frac{1}{d}(e_1 - e_2) I \end{pmatrix}.$$

Then the elastic free energy density can be written as

$$\psi_e = \frac{1}{2}\kappa \left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d}\mu (e_1 - e_2)^2 + \frac{1}{2}|y|^2 \mu |u_2'|^2.$$

We conclude that the elastic free energy per unit axial length is

$$\int_\Sigma \psi_e dy = \frac{1}{2} A\kappa \left(e_1 + (d-1)e_2\right)^2 + \frac{d-1}{d} A\mu (e_1 - e_2)^2 + \frac{1}{2} J\mu |u_2'|^2$$

where

$$A = \int_\Sigma dy \quad \text{and} \quad J = \int_\Sigma |y|^2 dy$$

are the polar and the polar second order moment of area.

2.2. **One dimensional kinematics.**

3. The tensile test at large deformations

4. Shear band localization in cohesive materials

Appendix A. Modeling in continuum mechanics

Appendix B. Tools from functional analysis

B.1. **Nonlinear analysis.**

Theorem B.1. Let V a Banach space and let $f \in C^1(V)$ and $K \subset V$ compact. Then f is Lipchitz continuous on K.

Proof. By contradiction let u_n and v_n two sequences in K such that
$$\frac{|F(u_n)-F(v_n)|}{\|u_n-v_n\|}\geq n.$$
By compactness of K up to extracting subsequences we can assume that $u_n\to u$ and $v_n\to v$. If $u\neq v$ then
$$\frac{|F(u_n)-F(v_n)|}{\|u_n-v_n\|}\to\frac{|F(u)-F(v)|}{\|u-v\|}\infty$$
which is a contradiction. If $u=v$ then
$$\frac{|F(u_n)-F(v_n)|}{\|u_n-v_n\|}=\frac{|\langle DF(u),u_n-v_n\rangle+o(\|u_n-v_n\|)|}{\|u_n-v_n\|}\leq\|DF(u)\|+o(1)\to\|DF(u)\|\leq\infty$$
which is a contradiction. □

B.2. **Convex analysis.**

Example B.2. Let $f(v)=\frac{1}{2}\|v\|^2$. Then $f^*(\xi)=\frac{1}{2}\|\xi\|^2$. Indeed
$$\langle\xi,v\rangle-\frac{1}{2}\|v\|^2\leq\frac{2\|\xi\|\|v\|-\|v\|^2}{2}\leq\frac{\|\xi\|^2}{2}.$$
This implies $f^*(\xi)\leq\frac{\|\xi\|^2}{2}$. To get the opposite inequality, fix $\epsilon>0$ and let $v\in V$ such that $\|v\|=\|\xi\|$ and $\langle\xi,v\rangle\geq\|\xi\|^2-\epsilon$. Then
$$f^*(\xi)\geq\|\xi\|^2-\epsilon-\frac{1}{2}\|\xi\|^2\geq\frac{1}{2}\|\xi\|^2-\epsilon.$$
By arbitrariness of ϵ we conclude.

Let f and g from V to $(-\infty,\infty]$. Their inf-convolution $f\Box g\colon V\to(-\infty,\infty]$ is
$$f\Box g(v)=\inf_{w\in V}(f(w)+g(v-w)).$$

Proposition B.3. (inf-convolution duality formula) Let f and g from V to $(-\infty,\infty]$. Then

(i) $(f\Box g)^*=f^*+g^*$
(ii) $(f+g)^*\leq f^*\Box g^*$
(iii) if there exists $v\in\operatorname{dom}f\cap\operatorname{dom}g$ and f is continuous at v then $(f+g)^*=f^*\Box g^*$.

Proof. Step 1. It holds that
$$\begin{aligned}(f\Box g)^*&=\sup_{v\in V}\left(\langle\xi,v\rangle-\inf_{w\in V}(f(w)+g(v-w))\right)=\\&=\sup_{v\in V}\sup_{w\in V}(\langle\xi,w\rangle-f(w)+\langle\xi,v-w\rangle-g(v-w))=\\&=\sup_{w\in V}\sup_{v\in V}(\langle\xi,w\rangle-f(w)+\langle\xi,v-w\rangle-g(v-w))=\\&=\sup_{w\in V}(\langle\xi,w\rangle-f(w)+g^*(\xi))=f^*(\xi)+g^*(\xi).\end{aligned}$$
This establish (i).

Step 2. By the Young–Fenchel inequality
$$f^*(\zeta)+g^*(\xi-\zeta)\geq\langle\xi,v\rangle-f(v)-g(v).$$
Taking the supremum over $v\in V$
$$f^*(\zeta)+g^*(\xi-\zeta)\geq(f+g)^*$$

and taking the infimum over $\zeta \in V$

$$f^* \Box g^* \geq (f+g)^*.$$

This proves (ii).

Step 3. Now we prove (iii). If $(f+g)^*(\xi) = \infty$ then (iii) is satisfied at ξ. Then suppose that $(f+g)^*(\xi) \in [-\infty, \infty)$. Since $\operatorname{dom} f \cap \operatorname{dom} g \neq \emptyset$ then actually $(f+g)^*(\xi) = c \in (-\infty, \infty)$. Let

$$A = \{(a,v) \in \mathbb{R} \times V \mid a \leq \langle \xi, v\rangle - g(v) - c\}.$$

Clearly A is nonempty and convex. We show that A and $\operatorname{int}(\operatorname{epi} f)$ are disjoint. Indeed if by contradiction $(a,v) \in A \cap \operatorname{int}(\operatorname{epi} f)$ then

$$f(v) < a \leq \langle \xi, v\rangle - g(v) - c$$

and

$$c < \langle \xi, v\rangle - f(v) - g(v) \leq (f+g)^*(\xi) = c.$$

Then by Hahn–Banach theorem there exists $(\gamma, \zeta) \in \mathbb{R} \times V^*$ such that

$$\gamma a + \langle \zeta, v\rangle \leq \gamma b + \langle \zeta, w\rangle \tag{B.1}$$

for all $(a,v) \in \operatorname{int}(\operatorname{epi} f)$ and $(b,w) \in A$. If we fix $v = w \in \operatorname{dom} f \cap \operatorname{dom} g$, $b \in \mathbb{R}$ sufficiently small such that $(b,v) \in A$ and we let $a \to \infty$, we discover that $\gamma \leq 0$ necessarily. If $\gamma = 0$ then inequality (B.1) and Hahn–Banach theorem imply that $\operatorname{dom} f$ and $\operatorname{dom} g$ are separated by a closed hyperplane. This in turn implies that $\operatorname{int}(\operatorname{dom} f)$ and $\operatorname{dom} g$ are disjoint, but this contradicts the hypotheses so that $\beta < 0$. In this case we set $\theta = |\beta|^{-1} f \zeta$ and from inequality (B.1) we get

$$\begin{aligned} f^*(\theta) &= \sup\{\langle \theta, v\rangle - f(v) \mid v \in V\} = \\ &= \sup\{\langle \theta, v\rangle - a \mid (a,v) \in \operatorname{epi} f\} \leq \\ &\leq \inf\{\langle \theta, v\rangle - a \mid (a,v) \in A\} = \\ &= c + \inf\{\langle \theta - \xi, v\rangle + g(v) \mid v \in V\} = c - g^*(\xi - \theta). \end{aligned}$$

Then

$$f^* \Box g^*(\xi) \leq f^*(\theta) + g^*(\xi - \theta) \leq c = (f+g)^*(\xi).$$

This concludes the proof. □

Proposition B.4 (properties of positively 1-homogeneous convex functions)**.** Let $f\colon V \to (-\infty, \infty]$ be a proper, l.s.c., positively 1-homogeneous and convex function. Then

(i) $f(0) = 0$ and f is subadditive
(ii) for each $x \in \operatorname{dom} f$ it holds $\partial f(x) = \{\xi \in \partial f(0) \mid f(x) = \langle \xi, x\rangle\}$
(iii) $f^* = \iota_K$ with $K = \partial f(0)$ and $f = \sigma_K$.

Proof. Step 1. Let $v \in \operatorname{dom} f$. Since f is l.s.c. and is positive 1-homogeneous

$$f(0) \leq \liminf_{n\to\infty} f\left(\frac{1}{n} v\right) = \liminf_{n\to\infty} \frac{f(v)}{n} = 0.$$

By positively 1-homogeneity and convexity we have

$$\frac{1}{2} f(v_0 + v_1) = f\left(\frac{v_0 + v_1}{2}\right) \leq \frac{f(v_0) + f(v_1)}{2}.$$

This proves (i).

Step 2. We prove that for all $\xi \in V^*$ it holds that $f^*(\xi) \in \{0, \infty\}$. Indeed let $\xi \in V^*$ such that $f^*(\xi) < \infty$. Then since f is positively 1-homogeneous, it holds $f^*(\xi) = 2^{-1} f^*(\xi)$ and this implies $f^*(\xi) = 0$.

Step 3. Let $v \in \operatorname{dom} f$ and $\xi \in \partial f(v)$. This is equivalent to the equality $f(v) + f^*(\xi) = \langle \xi, v\rangle$ which in turn implies that $f^*(\xi) < \infty$ and by previous step $f^*(\xi) = 0$. Then $f(v) = \langle \xi, v\rangle$ and for all $w \in V$

$$f(w) \geq f(v) + \langle \xi, w - v\rangle = \langle \xi, w\rangle.$$

This is equivalent to $\xi \in \partial f(0)$. We have proved that if $\xi \in \partial f(v)$ then $f(v) = \langle \xi, v\rangle$ and $v \in \partial f(0)$. It is immediate to check that also the converse is true. This proves (ii).

Step 4. It holds that $\xi \in K$ if and only if $f^*(\xi) = f(0) + f^*(\xi) = \langle \xi, 0\rangle = 0$. This proves that $f^* = \iota_K$. Since f is l.s.c. then $f = f^**= \iota_K^* = \sigma_K$. This establishes (iii). □

Appendix C. Balanced-viscosity solutions

(S) $\qquad V$ is a separable and reflexive Banach space

(D.1) $\qquad \mathcal{R}$ and $\mathcal{V}$ from V into $[0,\infty)$ are continuous, convex and strictly positive in $V \setminus \{0\}$

From Proposition B.4 it holds $\mathcal{R}(0) = 0$.

(D.2) $\qquad \mathcal{R}$ is positively 1-homogeneous

(D.3) $\qquad \mathcal{V}(v) = \phi(\|v\|)$ with $\phi \in C^1([0,\infty))$ convex such that $\phi(0) = 0$, $\phi'(0) = 0$, $\phi(r) > 0$ for $r > 0$ and $\lim_{r\to\infty} \phi'(r) = \infty$

We can extend ϕ to an even $C^1(\mathbb{R})$ function.

Let

$$\Phi(v) = \mathcal{R}(v) + \mathcal{V}(v)$$

and for $\eta > 0$

$$\Phi_\eta(v) = \eta^{-1}\Phi_1(\eta v) = \mathcal{R}(v) + \eta^{-1}\mathcal{V}(\eta v) = \mathcal{R}(v) + \mathcal{V}_\eta(v).$$

It holds that $\Phi_\eta^*(\xi) = \eta^{-1}\Phi^*(\xi)$. Since Φ_1 is convex and $\Phi_1(0) = 0$ then for all $v \in V$ the functions $\eta \mapsto \Phi_\eta(v)$ are increasing. Then it is well defined

$$\Phi_0(v) = \lim_{\eta\to 0^+} \Phi_\eta(v) = \inf_{\eta>0} \Phi_\eta(v) = \mathcal{R}(v).$$

Last inequality holds true since $\phi(0) = 0$ and $\phi'(0) = 0$.

Let $K = \partial\mathcal{R}(0)$. By Proposition B.4 it follows that $\mathcal{R} = \sigma_K$ and $\mathcal{R}^* = \iota_K$.

(E.1) $\qquad \mathcal{E} \colon D \times [0,T] \to \mathbb{R}$ with $D \subset V$

We set $\mathcal{E}(u,t) = \infty$ if $u \notin D$

(E.2) $\qquad \mathcal{G}(u) = \mathcal{R}(u) + \sup_{t\in[0,T]} \mathcal{E}(u,t)$ has compact sublevels $D_E = \{u \in D \mid \mathcal{G}(u) \leq E\}$ for all $E > 0$

(E.3) $\qquad$ regularity property of $\mathcal{E}$, I will find a simplified version of the properties stated in the article. For the moment I assume that $\mathcal{E}(u,t) = \Psi(u) + \mathcal{F}(t)$ with Ψ Fréchet differentiable and coercive and $\mathcal{F} \in C^1([0,T], V^*)$

Let $\mathcal{R}\colon V \to [0,\infty)$ satisfying (D.1) and (D.2). We say that a map $u\colon [0,T] \to V$ is $\mathrm{AC}_{\mathcal{R}}([0,T],V)$ if there exists $m \in L^1((0,T),[0,\infty))$ such that

$$\mathcal{R}(u(t_1)-u(t_0)) \le \int_{t_0}^{t_1} m(s)ds \quad \text{for every } 0 \le t_0 < t_1 < T.$$

Given a map $u\colon [0,T] \to V$ we define the total variation on $[a,b] \subset [0,T]$ as

$$\mathrm{Var}_{\mathcal{R}}(u,[a,b]) = \sup\left\{ \sum_{k=1}^{N} \mathcal{R}(u(t_k - t_{k-1})) \;\middle|\; \substack{N\in\mathbb{N} \\ a=t_0<\dots<t_N=b} \right\}.$$

We say that $u \in \mathrm{BV}_{\mathcal{R}}([0,T],V)$ if $\mathrm{Var}_{\mathcal{R}}(u,[0,T]) < \infty$. When $\mathcal{R} = \|\cdot\|$ we omit the subscript.

Lemma C.1. *It holds that $BV([0,T],V) \subset BV_{\mathcal{R}}([0,T],V)$.*

Proof. *Step 1.* We first prove that there exists $C > 0$ such that $\mathcal{R}(v) \le C\|v\|$. Indeed by contradiction suppose that there exists a sequence such that $\mathcal{R}(v_k) \ge k\|v_k\|$. By 1-homogeneity we can suppose that $\|v_k\| = 1$ without loss of generality. Moreover up to extracting a subsequence $v_k \rightharpoonup v$.

My proof is wrong, but in the case $V = L^2(\Omega)$ with Ω bounded and $\mathcal{R} = \|\cdot\|_1$ for example the result is trivially true.

Step 2. By previous step $\mathrm{Var}_{\mathcal{R}}([0,T],V) \le C\,\mathrm{Var}([0,T],V)$. This completes the proof. □

Proposition C.1. Let $u \in AC([0,T],V)$. Then $t \mapsto \mathcal{E}(u(t),t)$ is absolutely continuous in $[0,T]$ and

$$\frac{d}{dt}\mathcal{E}(u(t),t) = \langle \partial_u \mathcal{E}(u(t),t), \dot{u}(t)\rangle + \langle \dot{\mathcal{F}}(t), u(t)\rangle \quad \text{for a.e. } t \in (0,T).$$

Proof. *Step 1.* Since u is continuous then $u([0,T])$ is compact. Indeed let $u_n = u(t_n)$ be a sequence in $u([0,T])$. Then up to a subsequence $t_n \to t$ and by continuity $u_n = u(t_n) \to u(t)$.

Step 2. Since $\mathcal{E} \in C^1(V \times [0,T])$ then by Weierstrass theorem $\mathcal{E}$ and $\partial_t\mathcal{E}$ is bounded □

By the inf-convolution duality formula

$$\Phi_\epsilon^*(\xi) = (\mathcal{R}+\mathcal{V}_\eta)^*(\xi) = \inf_{\zeta\in V}\left(\mathcal{R}^*(\zeta) + \mathcal{V}_\eta^*(\xi-\zeta)\right) = \eta^{-1}\inf_{\zeta\in K}\mathcal{V}^*(\xi-\zeta) = \eta^{-1}\min_{\zeta\in K}\mathcal{V}^*(\xi-\zeta). \tag{C.1}$$

Last equality holds true because, since K is closed and convex, it is also weakly closed, $\mathcal{V}$ is convex and coercive, then the infimum is actually a minimum. Now $\mathcal{V}^*(\xi) = \phi^*(\|\xi\|)$.

Lemma C.2. *The function ϕ^* satisfies $\phi^*(0) = 0$, $(\phi^*)'(0) = 0$ and is monotone.*

Proof. Let $r = \rho(\gamma)$ the solution to $\gamma - \phi'(\rho(\gamma)) = 0$. Clearly $\rho \in C^1(\mathbb{R})$ and $\rho(0) = 0$. Then $\phi^*(\gamma) = \gamma\rho(\gamma) - \phi(\rho(\gamma))$. Now $\phi^*(0) = -\phi(0) = 0$ and $(\phi^*)'(0) = -\phi'(0) = 0$. □

Using that lemma and (C.1)

$$\Phi_\epsilon^*(\xi) = \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|). \tag{C.2}$$

The dissipation contact potential $\mathfrak{p}\colon V \times V^* \to [0,\infty)$ is

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\Phi_\eta(v) + \Phi_\eta^*(\xi)\right) = \inf_{\eta>0}\eta^{-1}\left(\Phi(\eta v) + \Phi^*(\xi)\right).$$

By (C.2)

$$\mathfrak{p}(v,\xi) = \inf_{\eta>0}\left(\mathcal{R}(v) + \eta^{-1}\phi(\eta\|v\|) + \eta^{-1}\min_{\zeta\in K}\phi^*(\|\xi-\zeta\|)\right) =$$
$$= \mathcal{R}(v) + \inf_{\eta>0}\left[\eta^{-1}\min_{\zeta\in K}\left(\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)\right)\right] =$$
$$= \mathcal{R}(v) + \min_{\zeta\in K}\inf_{\eta>0}\frac{\phi(\eta\|v\|) + \phi^*(\|\xi-\zeta\|)}{\eta}$$

where last equality holds because the minimum point does not depends on η. But for all r and γ in $\mathbb{R}$

$$\inf_{\eta>0}\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} = \gamma r.$$

Indeed by Young–Fenchel inequality

$$\frac{\phi(\eta r) + \phi^*(\gamma)}{\eta} \geq \gamma r$$

and the identity holds for η such that $\gamma = \phi'(\eta r)$. Then

$$\mathfrak{p}(v,\xi) = \mathcal{R}(v) + \|v\|\min_{\zeta\in K}\|\xi-\zeta\|. \tag{C.3}$$

Example C.2. Let $V = L^2(\Omega)$ with $\Omega\subset\mathbb{R}^d$ open and bounded. Let

$$\mathcal{R}(v) = \int_\Omega |v|dx \quad\text{and}\quad \mathcal{V}(v) = \frac{1}{2}\int_\Omega |v|^2 dx.$$

They satisfy the assumptions (D). In particular the continuity of $\mathcal{R}$ follows from Hölder inequality

$$|\mathcal{R}(v) - \mathcal{R}(w)| \leq \int_\Omega \left||v|-|w|\right| dx \leq \int_\Omega |v-w|dx \leq C\|v-w\|_2.$$

It holds that K is the closed unit ball in $L^\infty(\Omega)$. Indeed if $\xi\in V^* = L^2(\Omega)$ satisfy $\|\xi\|_\infty\leq 1$ then

$$\mathcal{R}^*(\xi) = \sup_{v\in V}\int_\Omega (\xi v - |v|)\,dx \leq \sup_{v\in V}\int_\Omega (\xi v - |v|)\,dx \leq \sup_{v\in V}\int_\Omega (|\xi|-1)|v|dx \leq 0$$

and $\mathcal{R}^*(\xi)\geq 0$ trivially. If $\|\xi\|_\infty > 1$ then there exists $\epsilon>0$ and $w\in L^1(\Omega)$ such that $\langle\xi,w\rangle\geq 1+\epsilon$. Since $L^2(\Omega)$ is dense in $L^1(\Omega)$ then for all $\delta>0$ there exist $v\in L^2(\Omega)$ such that $\|w-v\|\leq\delta$. Then for all $\lambda>0$

$$\mathcal{R}^*(\xi) \geq \langle\xi,\lambda v\rangle - \|\lambda v\|_1 \geq \lambda\left(\langle\xi,w\rangle - \|w\|_1 - \delta(1+\|\xi\|)\right) \geq \lambda\left(\epsilon - \delta(1+\|\xi\|)\right).$$

Choosing δ such that $\epsilon-\delta(1+\|\xi\|)>0$ and letting $\lambda\to\infty$ we get $\mathcal{R}(\xi)=\infty$. Since $\mathcal{V}^* = \frac{1}{2}\|\cdot\|_2^2$ and by (C.1)

$$\Phi_\epsilon^*(\xi) = \frac{1}{2\eta}\min_{\zeta\in K}\|\xi-\zeta\|_2^2 = \frac{1}{2\eta}\int_\Omega \min_{|\zeta|\leq 1}|\xi-\zeta|^2 dx.$$

By (C.3) it follows

$$\mathfrak{p}(v,\xi) = \|v\|_1 + \|v\|_2\min_{\zeta\in K}\|\xi-\zeta\|_2.$$

Appendix D. Energetic solutions

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